Lecture notes

REAL VARIABLES I

Fall 2026

Lecture notes will be added here as the semester progresses. Until a note is published, use the reading listed in the daily lecture log.

Planned unit map

Tentative.

Unit	Topics
I	Motivation, null sets, countability, and outer measure
II	σ-algebras, measures, and Carathéodory measurability
III	Measurable functions and the Lebesgue integral
IV	Integral limit theorems and uniform integrability
V	L^p spaces, inequalities, completeness, convergence, and density
VI	Product measures, Tonelli's theorem, and Fubini's theorem